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Supplementary Materials for

Women's connectivity in extreme networks

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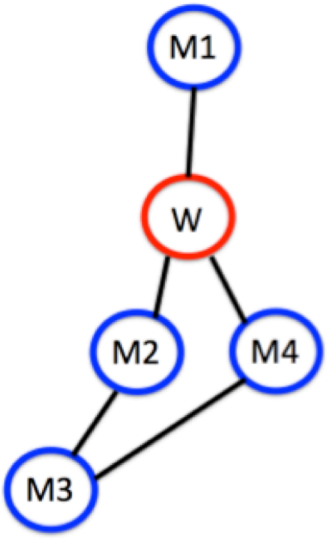
Supplementary Material (SM)



	M1	W	M2	M3	M4
M1	0	M1-W	M1-W-M2	M1-W-M2-M3 M1-W-M4-M3	M1-W--M4
W	W-M1	0	W-M2	W-M2-M3 W-M4-M3	W-M4
M2	M2-W-M1	M2-W	0	M2-M3	M2-W-M4 M2-M3-M4
M3	M3-M2-W-M1 M3-M4-W-M1	M3-M2-W M3-M4-W	M3-M2	0	M3-M4
M4	M4-W-M1	M4-W	M4-W-M2 M4-M3-M2	M4-M3	0

fig. S1 Importance of betweenness centrality (BC) in a network such as pro-ISIS or PIRA. Analysis of a representative network cluster to illustrate the specific importance of betweenness centrality in such extreme networks. For each pair of nodes chosen from the matrix, the shortest paths are shown. This is a common network motif, e.g. it happens to coincide with a worked example in COMS 998-006 Network Theory whose analysis we here verify independently and similar examples in Ref. 17 (see Ref. 17 of main text). The following analysis shows explicitly that an individual with high betweenness participates in a disproportionally large number of pathways for communications and brokerage. The example shown features 5 nodes where one represents a woman **W** and four men **M1**, **M2**, **M3**, **M4** and the matrix of shortest paths between these nodes. In particular, we find 20 paths out of 26 that involve the woman, 10 paths out of 26 that involve man M1, 14 paths out of 26 that involve man M2, 14 paths out of 26 that involve man M3 and 13 paths out of 26 that involve man M4. As a consequence, more shortest paths are available through the woman W than through any other node. If all pathways are used randomly, the woman W will be involved in the largest number making the woman W the most 'central' for communications, passing items and brokerage between distant parts of the

network. The same holds for the choice of node that is key for the stability of the network. For example, the number of paths between W and M3 is 2. If one node is removed, still another path between W and M3 exists. Even though M2 (or alternatively M4) is removed still a path would connect W and M3 through M4 (or M2), suggesting that neither M2 nor M4 are individually that topologically important. However, M1 remains disconnected if we remove W, making W's central location important. Betweenness Centrality (BC) formally captures this feature in a quantitative way. In particular, the BC of a node X measures the fraction of all shortest paths that pass through node X, indicating that a node with high BC has a big impact on the transfer of items through the network, under the assumption that efficient item transfer follows shortest paths. This feature is crucial for an extreme network such as PIRA or ISIS where items (e.g. information, material for IED) need to be passed quickly and efficiently since every extra step represents extra risk and potential cost. These practical constraints imply that an optimal path is potentially the shortest path. Applying the definition of BC to this example, the BC of W is (apart from a normalizing factor):

(fraction of paths from M1 to W through W) + (fraction of paths from M1 to M2 through W) + (fraction of paths from M1 to M3 through W) + (fraction of paths from M1 to M4 through W) + (fraction of paths from W to M2 through W) + (fraction of paths from W to M3 through W) + (fraction of paths from W to M4 through W) + (fraction of paths from M2 to M3 through W) + (fraction of paths from M2 to M4 through W) + (fraction of paths from M3 to M4 through W)

→ (0+1+0.5+0+1+0.5+0+1+1+0)/10=0.5

Analogously calculating the BC of M1-M4, we easily observe that BC(W) > BC(M1-M4).

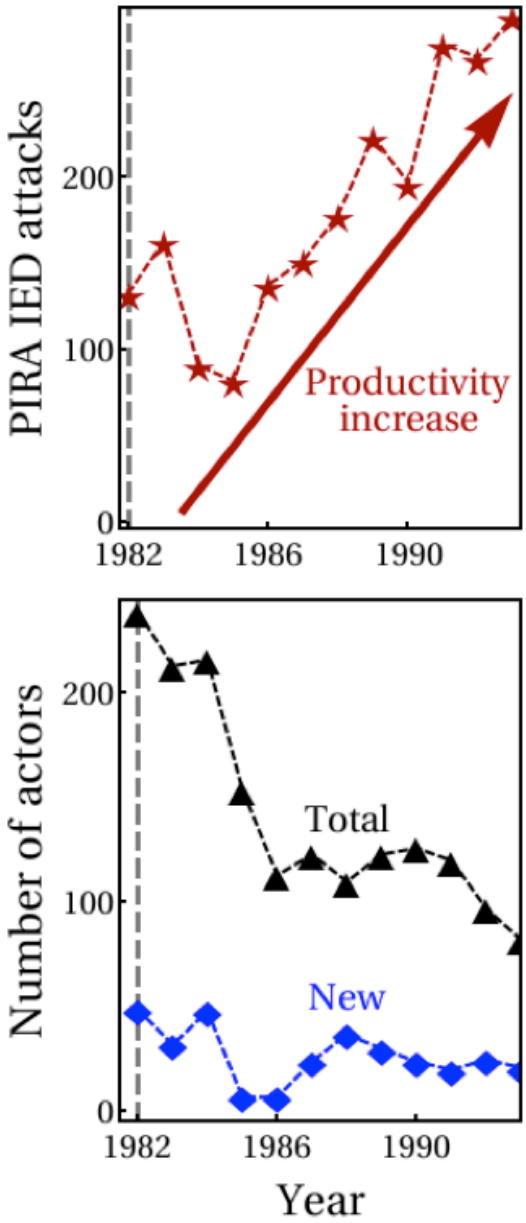


fig. S2. PIRA productivity not associated with increase in the number of PIRA actors. This figure indicates that the increase in PIRA productivity (i.e. number of attacks with IEDs) as shown in Fig. 2 in the main paper is not due to an increase in the number of PIRA actors. Indeed, the increase in the former seems to coincide with a decrease in total number of PIRA actors.

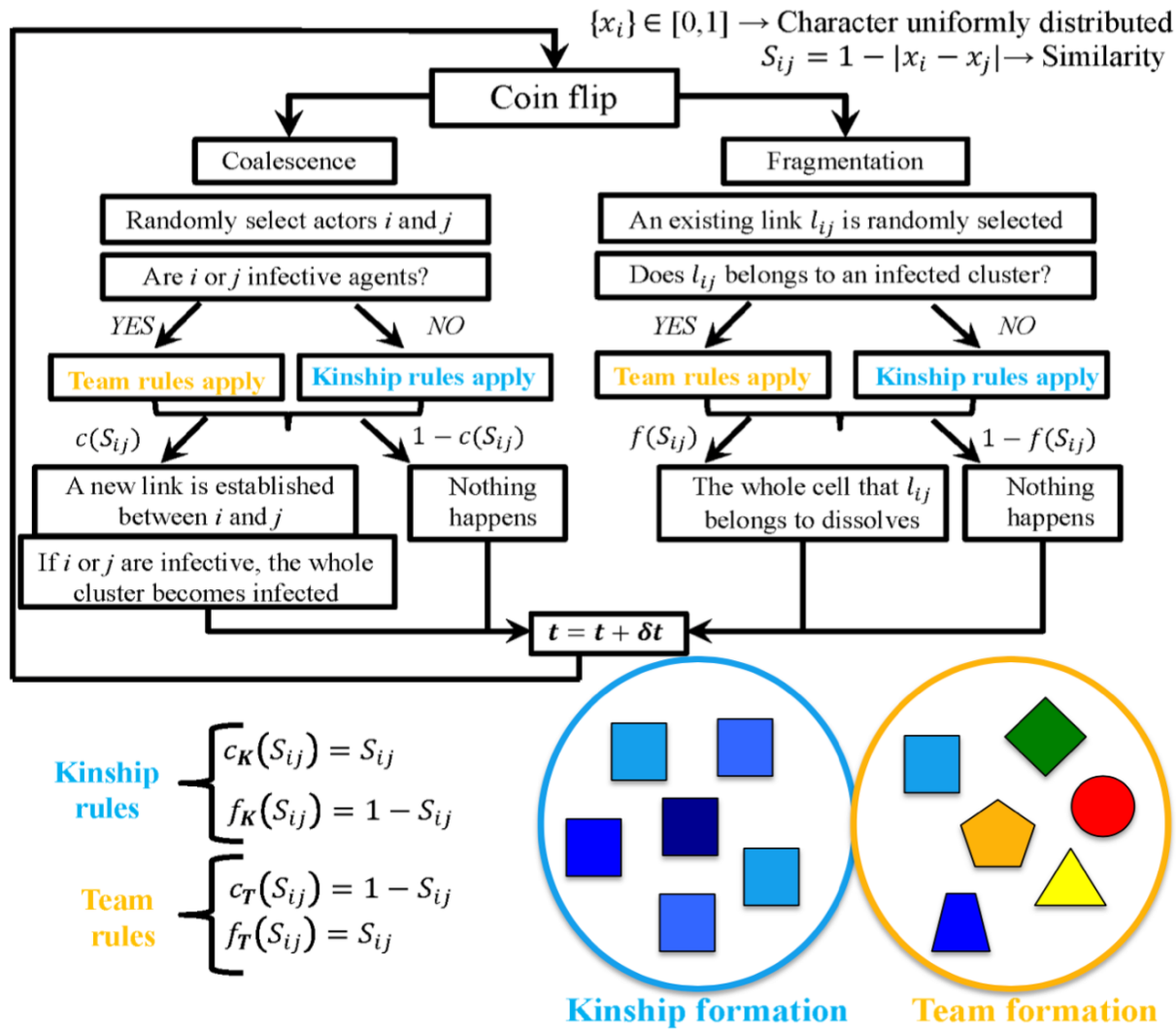


fig. S3. Schematic of our generative multi-agent model. This cartoon model, which we use solely to produce the results in Fig. 2E, makes several very significant assumptions; is oversimplified; and is very far from a definitive grouping mechanism for PIRA. On the plus side however, it (i) provides a minimal mechanism that generates results consistent with those observed in the actual PIRA network (see Fig. 2E); and (ii) is consistent with the notion that women might be more of a conduit than men for spreading information etc. because of their higher betweenness. Team-like operating structures are known to have developed at grass-roots level following the restructuring of PIRA between the end of the 1970's and beginning of 1980's. The model assumes N agents that are unconnected in the initial state since PIRA was at the start of its self-organization phase. The women are assumed to occupy better positions than men in the network (as indicated by their higher betweenness centrality) and are in this sense better spreaders. They spread the team ethic to others

with who they are in a cluster, like the spreading of a meme or infection. We assume that the men, by contrast, remain with a traditional kinship culture left over from PIRA's earlier days prior to reorganization. In each time step (δt ~1 day) a new connection may be created meaning that a cluster can grow, or an old cluster may be fragmented with equal probability. To model individual heterogeneity, each agent is labeled by a character x_i, a real number between *0* and *1* sampled from a uniform distribution. Susceptible (infected) agents follow kinship (team) formation rules that are defined in terms of the similarity $S_{ij} = 1 - |x_i - x_j|$ of actor *i* and *j*. Two highly similar agents *i* and *j* have $S_{ij} \sim 1$ since $| x_i - x_j | \sim 0$ and *vice versa*. In each time step, we randomly choose either the *coalescence* or *fragmentation* scenario. In the case of *coalescence* a pair of agents *i* and *j* are randomly picked. If one of the agents is infected *team formation* rules apply, linking actors with low similarity with probability of coalescence 1- S_{ij}. In turn, *kinship formation rules* apply linking actors with high similarity with probability of coalescence S_{ij}. Infected agents can spread the team ethic to the clusters they join, which is not the case for susceptible agents. In the case of *fragmentation*, an existing link l_{ij} between agents *i* and *j* is randomly picked. If l_{ij} is involved in an infected cluster, *team formation* rules apply, preferably dissolving all links in the underlying cluster if the similarity of *i* and *j*, S_{ij} is high. Otherwise, *kinship formation* rules apply, with a high probability of dissolving all links in the underlying cluster if similarity S_{ij} is low.

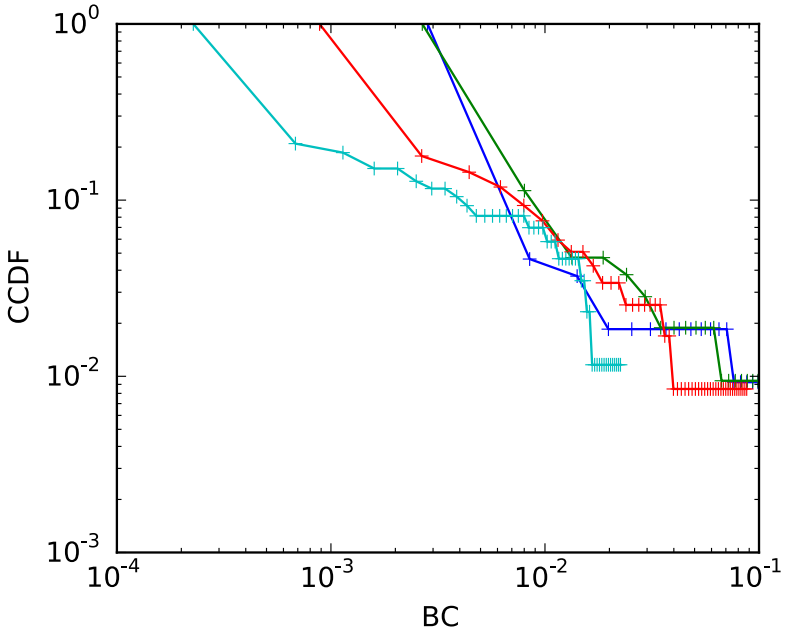


fig. S4. Evidence that the results of high betweenness centrality (BC) are not a consequence of a few exceptional women. The figure shows the complementary cumulative distribution function (CCDF) of betweenness centrality (BC) among the women in the online pro-ISIS network, obtained at four different representative timesteps. We note that there are 16,931 women and 24,883 men who are pro-ISIS followers online, indicating a high female to male ratio. The broadness of this distribution supports the notion that the high centrality does not emerge from a few special women within the subpopulation of 16,931 women. For PIRA, although their percentage is smaller, women with high betweenness centrality also do not appear to be exceptional in any way, nor are the centrality values dominated by a few women from the subset of all women. Indeed, most of these PIRA women are not even familiar names to subject matter experts. To double-check this latter point, we gave a randomly ordered list of PIRA women to subject-matter experts and asked them to rank the list in terms of relative importance. The result was that women with highest betweenness centrality did not rank high on the expert's lists (further details of this test are available from the authors). Such an observation confirms that women's high betweenness centrality is not a simple consequence of a few 'exceptional' women of PIRA, a result that we corroborated for ISIS women as well. Below we give four examples of high betweenness PIRA women to illustrate their non-exceptional nature. Since many of these women are still alive, we only use initials to identify them. None of these four women are notorious, or indeed are notorious by name to the subject matter experts that we asked. Their characteristics are unremarkable:

- *MC*: Age at recruitment 18. No university education. No notable other education. Not married. No children. On a more mundane level, a simple Google search of her name, with or without PIRA added to the search, shows no hits connected to her.
- *DR*: Age at recruitment 29. No university education. No notable other education. Not married. No children. Again, a simple Google search of her name, with or without PIRA, shows no hits connected to her.
- *HR*: Age at recruitment 70. No university education. No notable other education. Not married. No children. Again, a simple Google search of her name, with or without PIRA, shows no hits connected to her.
- *FM*: Age at recruitment unknown, which in part reflects the fact that so little is known about her. No university education. No notable other education. Not married. No children. Again, a simple Google search of her name, with or without PIRA, shows no hits connected to her.

More generally, we note that across the 1312 men and 70 women in our PIRA dataset, the variation in women's ages, backgrounds and locations is akin to the men's. This suggests that the women are also not collectively special in terms of their individual heterogeneity.

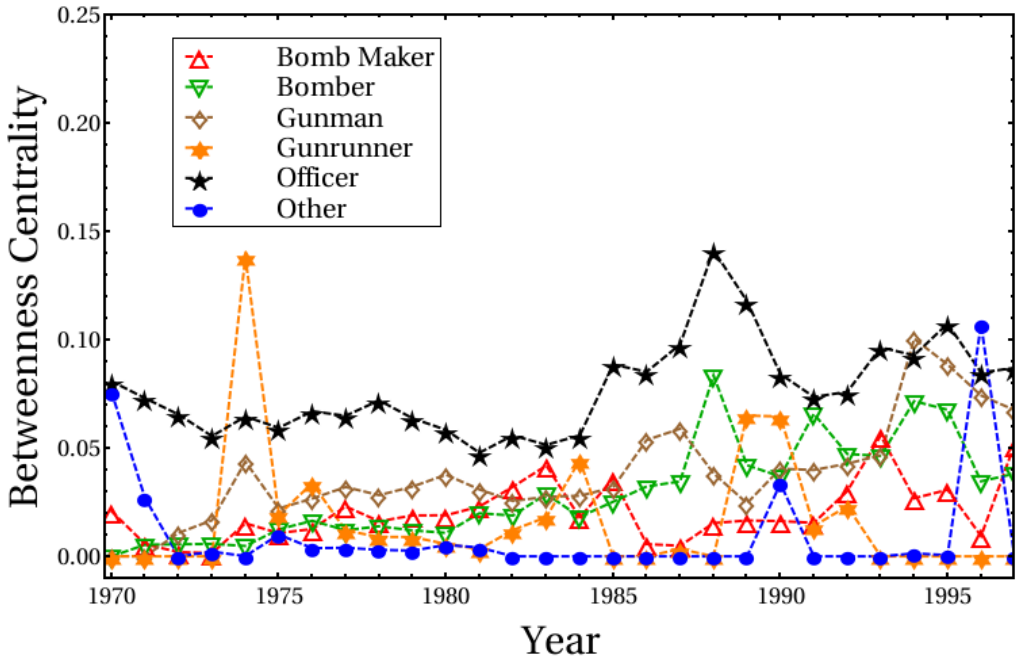


fig. S5. BC broken down by particular operational roles. The figure shows the normalized betweenness centrality (BC) broken down by an actor's speciality for PIRA. The original data show that both men and women are scattered across these roles. Hence it is unlikely that the difference in BC between men and women is coming from a particular partitioning of roles between men and women. This result, combined with the fact that there was a known bottom-up development of PIRA operational groups, suggests that women weren't simply being appointed to occupy particular roles that had high BC and that the high BC is more likely a bottom-up emergent phenomenon.